

The box model

In the last lesson, you added background colors to an article and a notice. The colors worked, but the boxes probably felt cramped because the text touched the edges. That is what this lesson fixes.

Every visible element on a page is drawn as a rectangular box. The box model is the way CSS describes the parts of that box: content, padding, border, and margin. Once you understand those four parts, spacing starts to feel much less random.

Start with a cramped box

If you still have the page from the last lesson, you can keep using it. If not, set up `index.html` and `style.css` for this box model exercise.

Use this HTML as the starting point:

html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Box model practice</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <main class="page">
      <article class="article-preview">
        <h1>Readable boxes need space</h1>
        <p>Background colors are useful, but they need room around the text to feel comfortable</p>
      </article>

      <p class="notice">Email is required before you can continue.</p>
      <p class="success-message">Profile saved successfully.</p>
    </main>
  </body>
</html>
```

Give the page a little color, but do not add any spacing yet:

CSS

```
body {
  font-family: system-ui, sans-serif;
  color: #111827;
  background-color: #f9fafb;
}

.page {
  max-width: 720px;
  margin: 40px auto;
}
```



```
.article-preview {  
  background-color: white;  
}  
  
.notice {  
  color: #92400e;  
  background-color: #fffbeb;  
}  
  
.success-message {  
  color: #166534;  
  background-color: #f0fdf4;  
}
```

Open the page. The backgrounds show up, but the text sits right against the edges. That is not a color problem; it is a box model problem.

Every element has a box

Even when you do not draw a border, the browser still treats each element as a box.

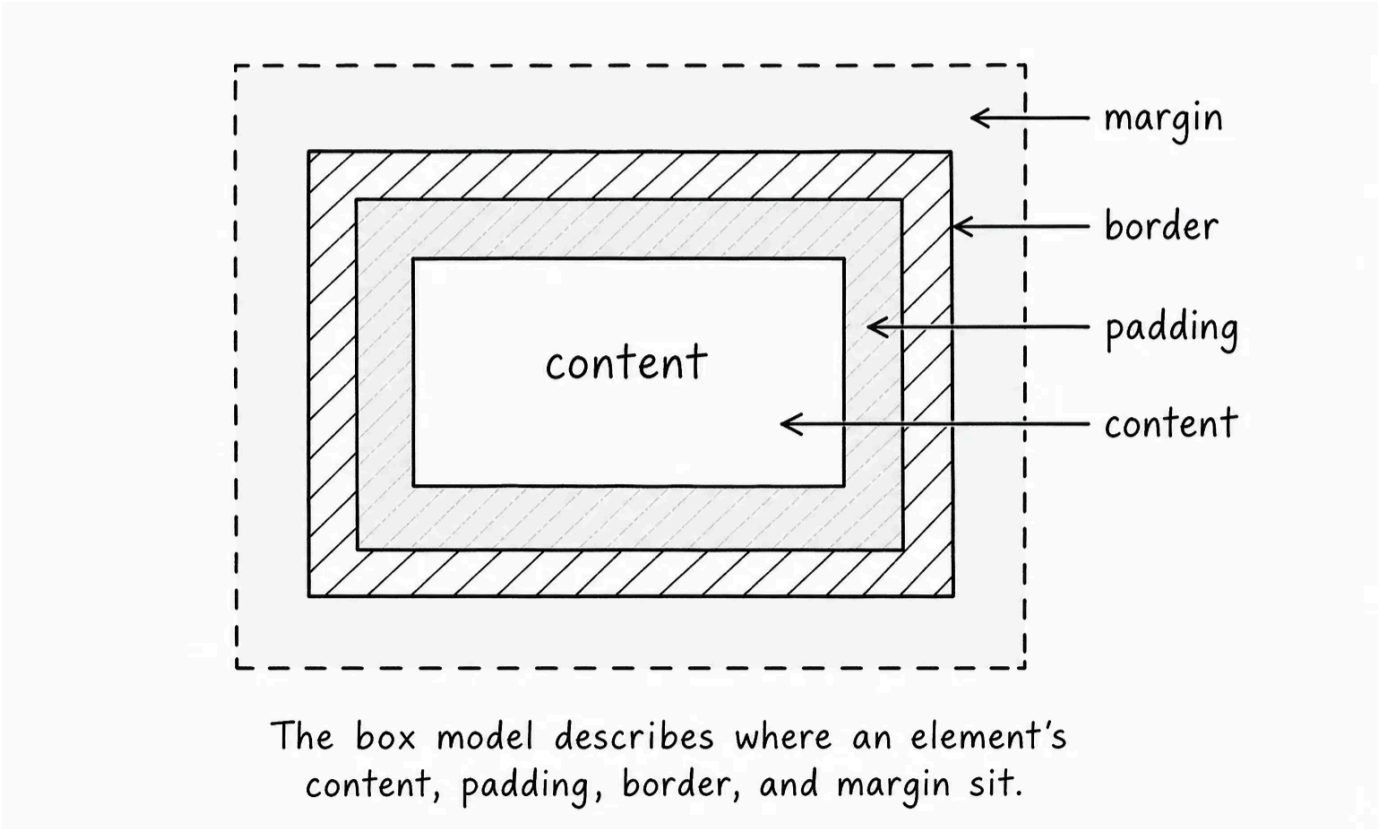
That box has four main parts:

- Content: the actual text, image, button label, or other content.

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- Border: a line around the padding and content.
- Margin: space outside the element, between this box and other boxes.



The important idea is simple: padding creates room inside the element, while margin creates room outside the element.

Add padding inside the box

Padding gives the content room inside the element. Add padding to the article and messages:

CSS

```
.article-preview {
  background-color: white;
  padding: 24px;
}

.notice {
  color: #92400e;
  background-color: #fffbeb;
  padding: 16px;
}

.success-message {
  color: #166534;
  background-color: #f0fdf4;
  padding: 16px;
}
```

Reload the page and the colored areas should feel less cramped because the text now has space around it.

There is a detail worth noticing: the element's background color fills the padded area too. That is why adding padding makes the colored notice feel larger, not just roomier. It grows the colored region itself, while margin creates space outside the colored region.

The `padding` property is a shorthand property. One value applies to all sides:

CSS

```
padding: 16px;
```

Two values are also common:

CSS

```
padding: 12px 16px;
```

The first value is top and bottom. The second value is left and right.

You can also target one side with properties like `padding-left` , `padding-right` , `padding-top` , or `padding-bottom` .

You do not need to memorize every shorthand form today. The main habit is this: use padding when the inside of a box needs more room.

Add a border around the box

A border is a line around the element. It can give a box a clearer edge, or it can add emphasis to one side.

The warning notice from the above HTML still relies mostly on color to signal that something needs attention. A left border can emphasize it without

changing the whole layout:

CSS

```
.notice {
  color: #92400e;
  background-color: #ffffeb;
  padding: 16px;
  border-left: 4px solid #f59e0b;
}
```

That colored stripe gives the notice a stronger warning shape.

You can also add a softer border around the article:

CSS

```
.article-preview {
  background-color: white;
  padding: 24px;
  border: 1px solid #e5e7eb;
}
```

The **border** shorthand usually includes three parts:

- Width: **1px**
- Style: **solid**
- Color: **#e5e7eb**

The **border** property targets all sides of the element. The side-specific version works the same way, but only for one side:

CSS

```
.notice {
  border-left: 4px solid #f59e0b;
}
```

Borders are useful when you want a box to have a clear edge. Later, you will learn related visual details like **border-radius** and **box-shadow** .

Add margin outside the box

Margin creates space outside an element.

Right now, the article and messages may sit too close together. Add margin between them:

CSS

```
.article-preview {
  background-color: white;
  padding: 24px;
  border: 1px solid #e5e7eb;
  margin-bottom: 24px;
}
```

```
}

.notice {
  color: #92400e;
  background-color: #ffffbe;
  padding: 16px;
  border-left: 4px solid #f59e0b;
  margin-bottom: 12px;
}
```

Reload the page and the boxes should now have space between them. Margin does not belong to the colored background because it sits outside the element.

That is the main difference to remember:

- Use padding when the content needs space **inside** its own background.
- Use margin when the whole element needs space **away from other elements**.

Margin has shorthand patterns too:

CSS

```
margin: 16px;
margin: 12px 16px;
margin-bottom: 24px;
```

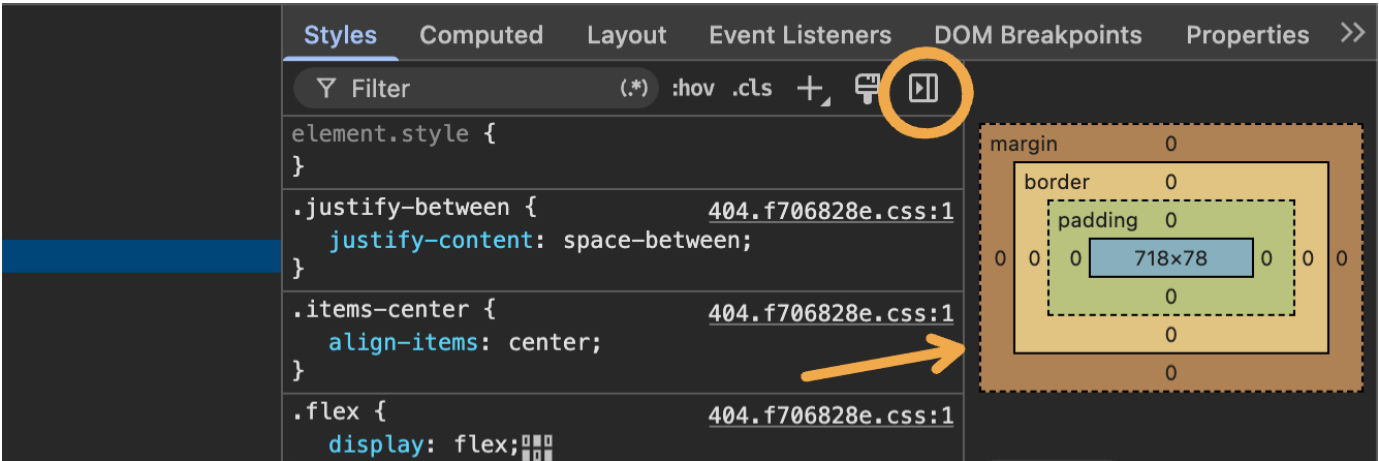
The pattern is similar to padding: one value applies to all sides, two values mean vertical then horizontal, and side-specific properties target one side.

 **Do not use margin for inner space** - If the text is touching the edge of a colored box, use `padding`, not `margin`. Margin moves the whole box away from other boxes, but it does not create space inside the background.

Inspect the box in DevTools

The box model is easier to understand when you inspect it in the browser.

Open DevTools, inspect the `.notice` element, and look for the box model diagram in the Styles or Computed panel. Browser DevTools vary a little, but the idea is the same: you can see the element's content, padding, border, and margin.



You will usually see a diagram with nested rectangles: content in the middle, then padding, border, and margin. Hover each layer if your browser supports it. The matching part of the element will highlight on the page.

Try changing the padding value in DevTools. The browser will update the page immediately.

Then try changing the margin value and notice that a different part of the page moves.

This is one of the best ways to debug spacing.

In the next lesson, you will see a related rule called **box-sizing: border-box**. It makes more sense once you start setting widths.

Try it

You have already made the colored boxes less cramped. Now use DevTools and push the box model a little further:

- Set **.notice** padding to **0**, reload, then set it back to **16px**.
- Change **.article-preview** padding from **24px** to **12px**, then to **32px**.
- Add **border-left: 4px solid #22c55e;** to **.success-message**.
- Change **.notice** **margin-bottom** from **12px** to **32px**.
- Remove the **border** from **.article-preview**, then add it back.
- Inspect **.notice** in DevTools and find its content, padding, border, and margin.

The goal is not to make the page beautiful yet. The goal is to know which part of the box you are changing.

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🕒 Knew

0 Items

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Test Your Understanding

What is the CSS box model?

[Click to Reveal the Answer](#)

- ✔ Already Know that
- ✖ Didn't Know that
- ⏩ Skip Question



LESSON COMPLETE

Nicely done.

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